

An End-to-End Reproducible Pipeline for 3D Multi-Organ CT Segmentation: From Data Curation to Fairness-Aware Hybrid Modeling

MedSegFM Workspace Study

April 19, 2026

Abstract

This paper presents a complete practical framework for 3D multi-organ CT segmentation developed in the MedSegFM workspace. The workflow spans dataset verification, deterministic data assembly and cleaning, standardized preprocessing, baseline model training, controlled variant comparisons, fairness enforcement, rich evaluation, failure analysis, and final hybrid model packaging. The core contribution is not a single architecture novelty, but a full reproducible experimentation system designed to reduce hidden confounders and support reliable model selection. The implemented stack includes configuration locking, fairness fingerprints, split-aware reporting, per-organ diagnostics, and failure-mode review outputs. The resulting system provides a robust foundation for full-scale training and external validation while preserving traceability of all decisions and artifacts.

1 Introduction

Automated 3D multi-organ segmentation in abdominal CT remains challenging due to anatomical variability, class imbalance, scanner heterogeneity, and annotation sparsity. While many studies report a final Dice score, fewer provide an end-to-end reproducible process that can be audited from raw data to packaged inference artifacts.

This work addresses that gap by building a complete research and engineering pipeline in a single workspace. The focus is methodological rigor: deterministic data handling, controlled comparisons, fairness validation across experimental variants, and clinically meaningful diagnostics beyond aggregate overlap metrics.

2 Objectives

The project was designed with five concrete objectives:

1. Establish a deterministic and auditable data pipeline from raw AMOS-style folders to train/val/test manifests.
2. Standardize spatial and intensity preprocessing for stable 3D model training.
3. Implement a modular baseline segmentation framework with configurable architecture and training options.
4. Enable fair comparisons across pretraining, adaptation, and prompting variants under locked settings.
5. Produce a final hybrid model workflow with quantitative comparison tables, failure analyses, and deployment-ready packaging.

3 Dataset and Data Governance

3.1 Raw Data Inspection

Initial inspection verified the distinction between the real dataset folder and archive metadata residue. The working dataset source was the AMOS directory, while the macOS extraction metadata directory was excluded from all training and evaluation logic.

3.2 Step 2: Dataset Assembly and Cleaning

A deterministic assembly stage was implemented to pair images and labels, validate shape consistency, detect missing pairs, and materialize split-specific manifests.

Observed run outcomes included:

- total discovered cases: 600,
- usable paired cases: 360,
- missing image-label pairs: 240,
- final split sizes: train 252, val 54, test 54.

This stage generated machine-readable manifests used as canonical split references for all downstream runs.

4 Standardized Preprocessing

4.1 Step 3 Pipeline

The preprocessing stage applied a consistent geometric and intensity normalization protocol:

- orientation normalization to RAS,
- resampling to fixed target spacing,
- intensity clipping and normalization for CT,
- crop/pad to fixed model input geometry,
- metadata export and quality reports.

4.2 Rationale

Without preprocessing standardization, model performance can vary due to scanner-specific spacing and intensity distributions rather than true modeling differences. This stage reduces nuisance variance and improves comparability between experiments.

5 Baseline Segmentation Framework

A modular package was implemented with clear separation of concerns for config, dataset loading, model construction, optimization, inference, and evaluation.

5.1 Core Design Principles

- Config-first reproducibility with explicit defaults and overrides.
- Deterministic behavior where feasible (seeded data flow and run metadata capture).
- CLI-driven orchestration for train, validate, infer, evaluate, and compare tasks.
- Structured experiment directories to preserve per-run state and outputs.

5.2 Implementation Notes

The framework supports baseline and transformer-based variants, with hooks for pretraining style, adaptation strategy, and prompting mode. This enabled a uniform training API for all experimental branches.

6 Controlled Experimental Axes

6.1 Pretraining Comparisons

The system supports scratch, supervised initialization, and self-supervised pretraining pathways. The primary goal is to compare initialization effects under fixed optimization and data settings.

6.2 Adaptation Comparisons

Adaptation strategies include full fine-tuning, partial decoder tuning, and parameter-efficient options. This enables analysis of quality-efficiency trade-offs.

6.3 Prompting Comparisons

Prompt-aware modes were integrated as controlled variants to evaluate whether additional contextual conditioning improves organ segmentation behavior.

7 Fairness and Reproducibility Controls

A fairness lock mechanism was implemented to prevent invalid comparisons caused by hidden configuration drift. Critical keys (seed, data split references, preprocessing ranges, optimization schedule, and training budget controls) are fingerprinted and checked across runs.

If variants violate locked settings, warnings are emitted and comparisons are marked as invalid or cautionary. This turns fairness from a documentation promise into an enforceable runtime rule.

8 Evaluation Protocol

8.1 Metrics

Evaluation was extended beyond mean Dice to include:

- Dice and IoU,
- HD95,
- precision and recall,
- volume error,
- surface-aware and calibration-related metrics when probability outputs are available.

8.2 Granularity

Reports are produced at multiple levels:

- per-case,
- per-organ,
- split summary,
- compact ranking tables for organ-wise strengths and weaknesses.

This design helps identify clinically relevant weak organs that would be masked by a single aggregate score.

9 Failure Analysis and Diagnostic Review

A dedicated failure-analysis module categorizes poor cases and patterns, including missed organs, over-segmentation, boundary inaccuracies, and structural inconsistencies. Representative qualitative panels are generated to support manual inspection.

This analysis layer transforms evaluation from passive scoring into actionable debugging intelligence for subsequent model refinement.

10 Final Hybrid Model Workflow

A final-hybrid pipeline was implemented to select strong components from comparison outputs, compose a final configuration, evaluate against baseline references, and package deployment artifacts.

Generated final artifacts include:

- final selection JSON,
- comparison and ablation CSV/JSON tables,
- packaged checkpoint and frozen config,
- reusable inference launchers.

11 Observed Quantitative Outcomes

On the latest smoke-scale evaluation, representative metrics included:

- final-hybrid mean Dice: 0.004281,
- baseline mean Dice: 0.004014,
- observed delta: +0.000267.

These values validate the end-to-end mechanics but should not be interpreted as definitive scientific performance due to smoke-scale sample size and sparse class coverage.

12 Qualitative Visual Results

Figure-based inspection complements scalar metrics by showing where boundaries are captured well and where organ confusion or under-segmentation appears.

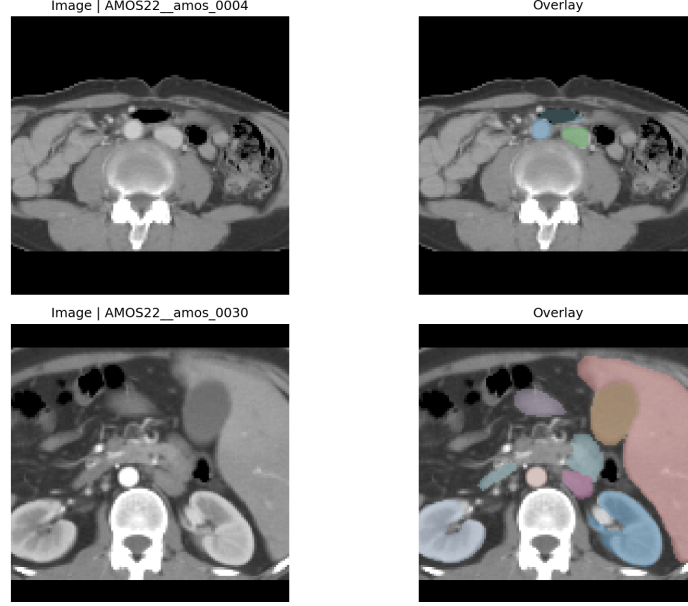


Figure 1: Preprocessing sanity check from standardized Step 3 outputs.

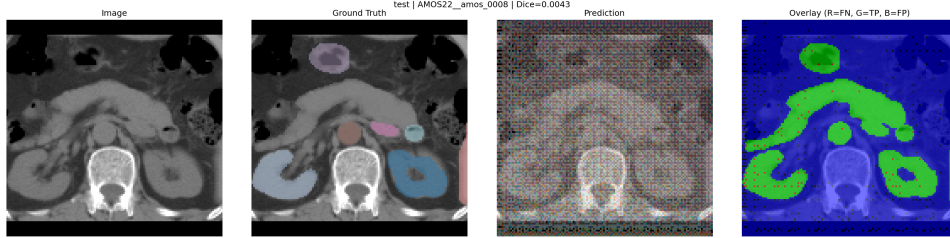


Figure 2: Final hybrid evaluation panel showing image, label, prediction, and error overlay for a representative test case.

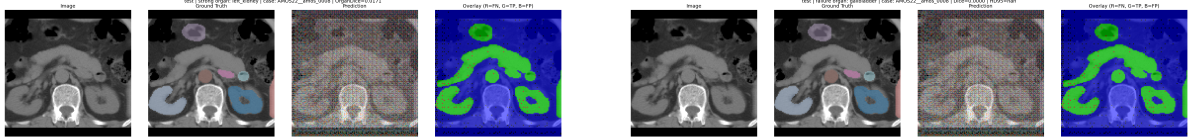


Figure 3: Organ-focused qualitative comparison: a stronger left-kidney segmentation (left) and a failure-case gallbladder segmentation (right).

13 Engineering Issues Encountered and Resolved

13.1 Windows Multiprocessing Pickling Constraint

A nested DataLoader worker seed function caused Windows spawn-mode pickling failures. The fix moved worker initialization to module scope, restoring training stability.

13.2 Inference Batch-Type Robustness

Batch collation edge cases produced type mismatches during inference. Input normalization logic was hardened to support heterogeneous batch structures.

13.3 NaN-Safe Aggregation

Small-sample evaluation occasionally produced undefined metrics. Finite-safe reducers were added to avoid unstable summary behavior.

14 Threats to Validity

- Smoke-scale runs are suitable for pipeline validation, not performance claims.
- Some metrics (for example HD95) can be undefined in sparse test scenarios.
- External generalization remains unverified until cross-dataset evaluation is executed.

15 Future Work

1. Execute full-scale training with strict fairness lock and no smoke overrides.
2. Perform external validation with explicit image/ground-truth directory routing.
3. Recompute final and ablation tables on full-size cohorts only.
4. Extend uncertainty-driven triage for clinically risky predictions.

16 Conclusion

This work delivers a complete, reproducible research pipeline for 3D multi-organ CT segmentation, spanning raw data governance to packaged final inference assets. The principal achievement is a fairness-aware experimentation framework that enables reliable model comparison, rich diagnostics, and transparent traceability. The system is ready for full-scale training and rigorous external validation studies.